

Aiden Mark

Machine Learning Engineer

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SUMMARY

Senior ML Engineer with 7+ years of experience delivering end-to-end machine learning solutions, including large language models (LLMs), agentic AI systems, deep learning, NLP, computer vision, and predictive analytics. Skilled in designing and deploying **scalable ML pipelines and data products**, operationalizing complex models, and developing intelligent automation solutions that produce measurable impact. Experienced in **leading teams, mentoring engineers, and collaborating with stakeholders** to translate data into actionable insights and strategic business value.

EXPERIENCE

Senior ML Engineer

Careviso

11/2023 - Present

- Built ML pipelines for data ingestion, training, and deployment. Ensured reproducible experiments, reliable performance, and scalability.
- Designed and deployed AI solutions including RAG-based workflows with LangChain and LangGraph, enabling scalable and intelligent applications.
- Built NLP-driven document parsing pipelines based on LLMs and NER models for prescriptions, medical bills, invoices, and clinical notes, enabling accurate and compliant extraction of critical patient and billing data.
- Developed and deployed predictive models, including predictive maintenance solutions that reduced equipment downtime resulting in significant operational savings.
- Designed and implemented AI-powered voice and conversational agents to deliver personalized health recommendations, automate surveys, and enhance patient engagement.
- Created solutions for Zoom call ingestion and automated transcription, providing fast, accurate extraction of patient and provider insights.
- Streamlined model deployment and orchestration using Docker, Kubernetes, CI/CD (GitLab), and cloud platforms (AWS, Azure), maintaining production reliability and scalability.
- Led ML lifecycle and MLOps architectures, mentored engineers, enforced engineering best practices, aligned deliverables with requirements.

Data Scientist

Mademarket

07/2020 - 11/2023

- Designed scalable data science pipelines for data ingestion, cleaning, and transformation, ensuring reproducible research, consistent model performance, and reliable production deployment.
- Led NLP initiatives to automate document parsing and entity recognition to extract structured data from unstructured financial documents.
- Implemented sentiment analysis models to extract insights from client communications, enhancing relationship management and enabling proactive engagement strategies.
- Designed and deployed a data-driven recommendation system using collaborative filtering and deep learning, increasing user engagement and improving deal discovery in investment banking CRM.
- Built and deployed a voice-based AI chatbot and call center assistant, improving customer support efficiency and reducing response times through natural language interactions.
- Developed an AI assistant powered by a centralized knowledge base, enabling bankers to quickly access insights, documents, and deal-related intelligence.
- Designed RESTful APIs to serve model outputs and insights, ensuring seamless integration into both internal tools and client-facing platforms with decoupled model and business logic.

Software Engineer

Cisco Systems Inc.

05/2018 - 07/2020

- Assisted in data preprocessing and feature engineering, including cleaning, encoding, imputation, transformation, scaling, and dimensionality reduction to prepare datasets for modeling.
- Built and optimized models based on PyTorch and TensorFlow, achieving state-of-the-art results.
- Supported the development of machine learning models for classification and regression tasks, applying deep learning techniques and implementing natural language processing models such as RNNs and LSTMs.
- Collaborated with team members to ensure reproducible experiments, clean code, and deployment-ready models, gaining hands-on experience in practical machine learning applications.

SKILLS

Programming & Frameworks

Python, PyTorch, TensorFlow, Keras, Scikit-Learn, Django, FastAPI, Pandas, NumPy, SciPy, PySpark, Spark MLlib, LangChain, LangGraph, N8N, Voice AI, Dask

Deployment & DevOps | Visualization & Monitoring:

Docker, Kubernetes, CI/CD (GitLab, Jenkins), Model Deployment & Monitoring, RESTful API Development, Cloud Platforms (AWS, Azure, GCP), Grafana, Tableau, Power BI, Matplotlib, Seaborn, Model Drift & Data Drift Monitoring, Dashboard Design for ML Operations

Machine Learning & AI

Supervised & Unsupervised Learning, Deep Learning (CNNs, RNNs, Transformers), Natural Language Processing (NLP), Sentiment Analysis, Large Language Models (LLMs), Agentic AI, Computer Vision & Image Classification, Predictive Modeling & Forecasting, Recommendation Systems (Collaborative Filtering, Deep Learning), Reinforcement Learning, Knowledge Graphs, Retrieval-Augmented Generation (RAG)

Data Engineering & Pipelines

Data Wrangling & Feature Engineering, Exploratory Data Analysis (EDA), ETL Pipelines, Data Ingestion & Transformation, Airflow, Databricks, SQL & NoSQL (Elasticsearch, MongoDB, PostgreSQL), Data Versioning & Validation

PROJECTS

1. HealthVision AI Suite

<https://www.careviso.com/>

HealthVision AI Suite is an end-to-end healthcare AI platform designed to automate the extraction and analysis of medical data. It combines NLP and computer vision pipelines to process unstructured medical documents, including prescriptions, bills, and clinical notes, and integrates real-time voice-based conversational agents for personalized patient interactions. The platform also features automated Zoom call transcription to capture clinical interactions, providing actionable insights for patient care and operational efficiency.

2. FinPulse

<https://www.mademarket.com/>

FinPulse is an AI-driven analytics platform for investment banking CRM systems that enhances client engagement and decision-making. The platform applies sentiment analysis on client communications, predictive modeling for relationship scoring, and forecasting for deal pipelines. An AI assistant powered by a centralized knowledge base delivers instant insights to bankers, enabling smarter, faster decision-making across client portfolios. FinPulse transforms raw CRM data into actionable intelligence to optimize business outcomes.